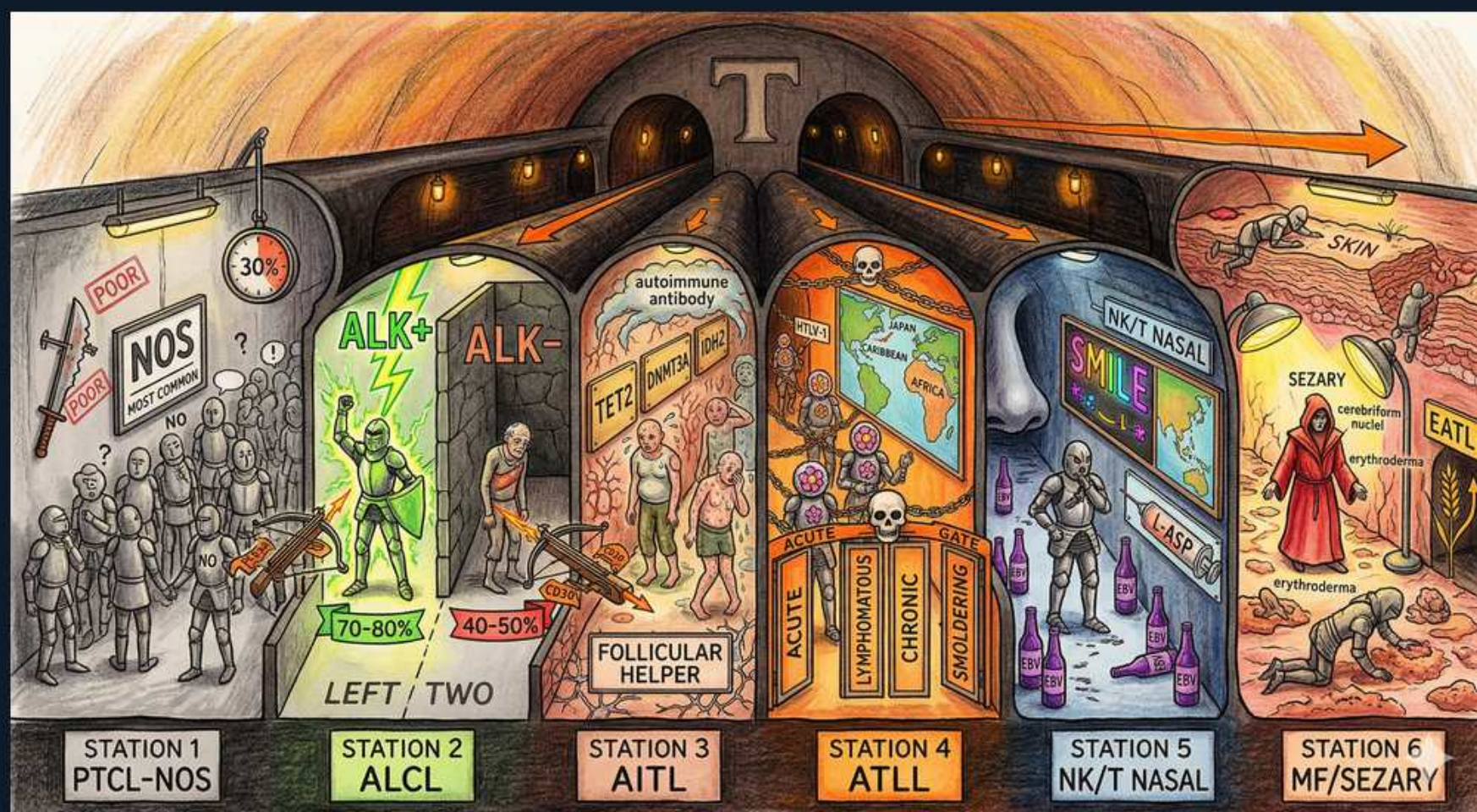


Mature T-Cell Lymphomas — The Subway System



1. PTCL-NOS (Station 1)

WHAT YOU SEE

Grey crowd of generic figures, 'NOS' sign, POOR banner, 30% bubble

WHAT IT ENCODES

Peripheral T-cell lymphoma, not otherwise specified is the most common mature T-cell lymphoma in the West (~30%). Diagnosis of exclusion after ruling out specific entities. Aggressive with poor prognosis; CHOP/CHOEP first-line, 5-yr OS only ~30%.

BOARD TRAP

Don't call it ALCL — PTCL-NOS lacks CD30 strong/uniform positivity and the hallmark cells; it is the leftover bucket.

2. ALCL ALK+ (Station 2)

WHAT YOU SEE

Green glowing figure, 'ALK+' green text, 70-80% banner

WHAT IT ENCODES

Anaplastic large cell lymphoma, ALK-positive carries t(2;5) NPM-ALK translocation. CD30+ always. Occurs in young patients; FAVORABLE prognosis, 5-yr OS 70-80%. Hallmark cells with horseshoe nuclei.

BOARD TRAP

ALK-positivity = good prognosis in ALCL. Don't confuse with ALK in lung cancer. ALK-negative ALCL (40-50%) is worse and DUSP22 rearrangement subset behaves like ALK+.

3. ALCL ALK- (Station 2)

WHAT YOU SEE

Orange 'ALK-' text, weaker figure, 40-50% banner

WHAT IT ENCODES

ALK-negative ALCL is CD30+ but lacks ALK translocation; older patients, intermediate prognosis (5-yr OS 40-50%). Systemic (not primary cutaneous) form. Brentuximab vedotin (anti-CD30) highly active.

BOARD TRAP

Primary cutaneous ALCL is a separate indolent skin-limited entity — don't treat it like systemic ALK- ALCL with aggressive chemo.

4. AITL (Station 3)

WHAT YOU SEE

Autoimmune antibody label, TFH/follicular helper text, FOLLICULAR HELPER

WHAT IT ENCODES

Angioimmunoblastic T-cell lymphoma arises from T-follicular-helper (TFH) cells. Classic triad: generalized lymphadenopathy, hypergammaglobulinemia, autoimmune phenomena (Coombs+ hemolysis, rash). EBV-driven B-cell expansion in background. CD10+, BCL6+, PD1+, CXCL13+.

BOARD TRAP

The polyclonal hypergammaglobulinemia and autoimmune features can mimic infection or autoimmune disease — biopsy shows arborizing high endothelial venules and TFH markers.

5. ATLL (Station 4)

WHAT YOU SEE

Skull, acute/lymphomatous/chronic/smoldering subtypes, Japan/Caribbean map

WHAT IT ENCODES

Adult T-cell leukemia/lymphoma is caused by HTLV-1 (endemic Japan, Caribbean). Four subtypes: acute, lymphomatous, chronic, smoldering. Acute/lymphomatous are aggressive. Hypercalcemia with lytic bone lesions, flower cells, CD4+CD25+.

BOARD TRAP

Hypercalcemia + lytic lesions can mimic myeloma — but ATLL has circulating flower cells and HTLV-1 serology. Strongyloides hyperinfection risk.

6. NK/T-Cell Nasal (Station 5)

WHAT YOU SEE

NK/T NASAL sign, SMILE regimen, nasal/midface figure, L-ASP bottles

WHAT IT ENCODES

Extranodal NK/T-cell lymphoma, nasal type is EBV-driven, angiocentric/angiodestructive causing midfacial destruction (lethal midline granuloma). CD56+, cytoplasmic CD3+. Treated with L-asparaginase-based SMILE regimen plus radiation.

BOARD TRAP

Don't mistake nasal destruction for granulomatosis with polyangiitis or invasive fungal sinusitis — EBER in-situ hybridization is positive in NK/T lymphoma.

7. Mycosis Fungoides (Station 6)

WHAT YOU SEE

Skin patches/plaques, SKIN label, cerebriform nuclei reference

WHAT IT ENCODES

Mycosis fungoides is the most common cutaneous T-cell lymphoma — indolent, skin-limited patches/plaques/tumors. CD4+ T-cells with cerebriform (Sezary/Lutzner) nuclei, epidermotropism with Pautrier microabscesses. Skin-directed therapy early.

BOARD TRAP

Early patch-stage MF mimics eczema/psoriasis for years — repeat biopsies often needed. Loss of CD7 supports clonal T-cell process.

8. Sezary Syndrome (Station 6)

WHAT YOU SEE

Red-robed figure, SEZARY, erythroderma, blood involvement

WHAT IT ENCODES

Sezary syndrome is the leukemic/erythrodermic variant of CTCL: triad of erythroderma, lymphadenopathy, and ≥1000/μL circulating Sezary cells. Worse prognosis than MF. CD4+CD7- clone in blood; treat systemically.

BOARD TRAP

Erythroderma has many causes (drug, psoriasis, atopic) — Sezary requires demonstrating the clonal blood population, not just red skin.

9. EATL

WHAT YOU SEE

EATL label near intestinal/gut region

WHAT IT ENCODES

Enteropathy-associated T-cell lymphoma is a gut T-cell lymphoma arising in refractory celiac disease (Type I). Presents with abdominal pain, perforation, obstruction. Poor prognosis. Monomorphic epitheliotropic intestinal TCL is the Type II (non-celiac) variant.

BOARD TRAP

In a celiac patient with new abdominal symptoms or refractoriness, think EATL — not just a celiac flare.

10. CD30 / Brentuximab Target

WHAT YOU SEE

ALK+/ALK- ALCL stations linked by CD30 biology

WHAT IT ENCODES

CD30 is the unifying target across ALCL (and CD30+ PTCL, MF). Brentuximab vedotin is an anti-CD30 antibody-drug conjugate (MMAE payload). CHOP+brentuximab (ECHELON-2) improved survival in CD30+ PTCL.

BOARD TRAP

Brentuximab works on CD30 expression level — not all PTCL-NOS express enough CD30 to benefit, unlike uniformly CD30+ ALCL.

11. T-Cell Origin (Tunnel)

WHAT YOU SEE

Large 'T' arch over the subway entrance, orange directional arrow

WHAT IT ENCODES

The whole 'underground' represents mature/peripheral T-cell and NK-cell neoplasms — post-thymic. Collectively ~10-15% of NHL, more aggressive than B-cell counterparts overall except indolent cutaneous forms.

BOARD TRAP

These are MATURE (peripheral) T-cell lymphomas — distinct from T-lymphoblastic lymphoma/T-ALL, which is a precursor (TdT+) neoplasm of thymic origin.
